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Fireblocks cloud architecture

Updated 5 months ago

Overview

Multiple providers and components make up the Fireblocks cloud architecture.

Service providers

Fireblocks uses three cloud service providers: Microsoft Azure, Amazon AWS, and Google Cloud Platform (Firebase).

Microsoft Azure

- Hosts core components used to handle and store sensitive information, such as key share material, configurations, policy rules, and third-party API credentials
- Securely stores data using [SGX Confidential Enclaves](#)

Amazon AWS

- Hosts various gateway and frontend services
- Does not store any secret cryptographic material, such as API keys and MPC key shares
- Orchestrates messaging between frontend and core components

Google Cloud Platform (Firebase)

- Acts as the caching database for the Fireblocks Console and the Fireblocks mobile app
- Enables efficient data delivery

Fireblocks provides a highly available (active-active and active-passive) cloud infrastructure by utilizing multiple Tier 1 cloud providers. Two co-signers are hosted on the Fireblocks SaaS cloud environment. The third co-signer is either your mobile device or a self-hosted server (in your cloud environment or your on-premises data center). It is suggested to

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System components

Fireblocks uses several components in the Fireblocks cloud architecture.

Shell Services (API gateways, event orchestration)

- Consists of frontend services, API Gateways, and message queues
- Orchestrates events between client applications and core components
- Does not store or manage sensitive data

Transaction Signing Modules (Co-signers)

- Securely stores MPC private key shares and signing transactions
- Integrates with your core services using a programmable interface
- Stores and manages sensitive data

Core Components

- Executes core Fireblocks services and modules
- Stores and manages sensitive data

Trusted Shared Services

- Executes shared core Fireblocks services and modules, including the Fireblocks P2P Network
- Stores and manages sensitive data

Blockchain Nodes Infrastructure

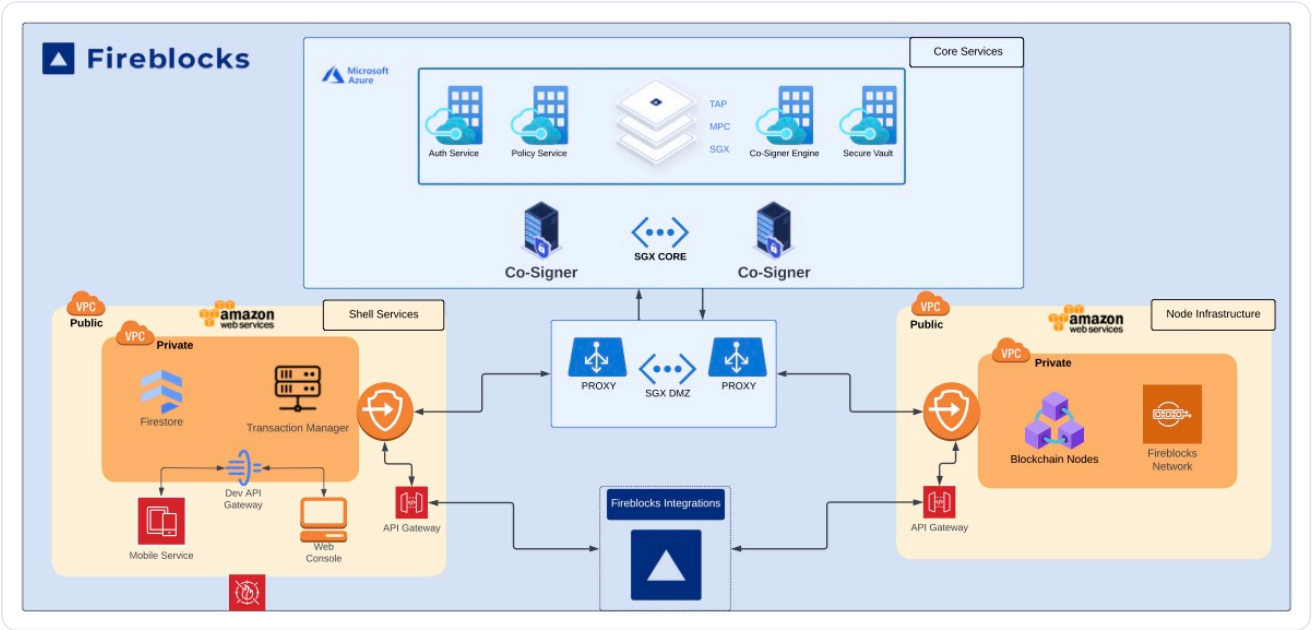
- Connects various supported blockchain networks and broadcasts signed messages
- Consists of both public and private (if any) networks
- Easily scales vertically and horizontally, and contains various code optimizations to provide efficient execution
- Does not store or manage sensitive data

Disaster Recovery Services

- Recovers all your assets in case of disaster
- Provides reconstruction of the extended ECDSA and EdDSA private keys (xprv+fprv)
- Should be stored on an offline air-gapped machine with hardened access permissions

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Deployment options



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